

Employee Management System

Week 01 – Full Stack Development Internship Task

Project Overview

The Employee Management System is a full-stack web application developed to manage employee records efficiently. It includes secure authentication, CRUD operations, responsive UI, and MySQL database integration.

Objective

Develop a complete full-stack web application demonstrating authentication, CRUD operations, database integration, backend API development, and a responsive interface.

Technologies Used

Frontend: HTML5, CSS3, Bootstrap 5, JavaScript

Backend: Node.js, Express.js

Database: MySQL

Packages: Express, MySQL2, bcrypt, JWT, CORS, dotenv, nodemon

Features

User Registration/Login, Password Hashing, JWT Authentication, Protected Routes, Employee CRUD, Dashboard, Search, Responsive Design.

Database Structure

Users: ID, Name, Email, Password, Created At

Employees: ID, Name, Email, Phone, Department, Position, Salary, Joining Date, Created At

API Endpoints

POST /api/auth/register

POST /api/auth/login

POST /api/auth/logout

GET /api/employees

POST /api/employees

PUT /api/employees/:id

DELETE /api/employees/:id

Security

Passwords are hashed using bcrypt. JWT is used for authentication and protected routes. Input validation is applied on both frontend and backend.

Conclusion

This project successfully fulfills the internship requirements by implementing a complete full-stack Employee Management System with authentication, CRUD operations, responsive design, backend integration, and MySQL database connectivity.